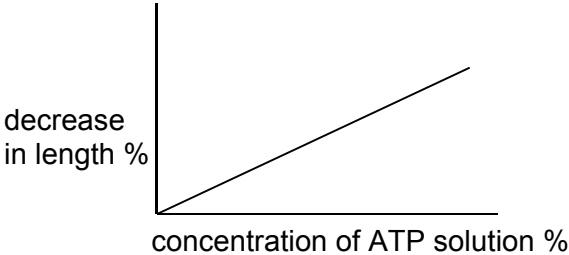


1(a)(i)	<i>independent:</i> type of (Ringer's) the solution ; <i>dependent :</i> (change in) length of muscle, strip / fibre / tissue AW ;	2
1(a)(ii)	<i>idea of:</i> the muscle fibres are different (starting) lengths ;	1
1(a)(iii)	to act as a <u>control</u> ; to show that Ringer's solution (alone) cannot cause contraction AW / to show that ATP is responsible for the contraction AW ;	2

Question	Answer	Marks
1(b)	<i>six from:</i> 1 ref. to a method of diluting the 0.5% ATP solution (with Ringer's solution) and to give at least 5 dilutions ; 2 ref. to at least 3 concentrations from 0.5% downwards with % units; 3 ref. to <u>control</u> using Ringer's solution (alone) ; 4 ref. to method for measuring change in length of fibres ; 5 ref. to using the <u>same</u> number fibres / strips for each concentration ; 6 ref. to adding the same volume <u>of ATP solutions</u> for each concentration ; 7 ref. to suitable volume of ATP solutions on a slide ; 8 ref. to leaving all fibres for the same / fixed (stated) time ; 9 ref. to low risk investigation / hazard <u>and</u> suitable safety precaution ; 10 ref. to <u>replicates</u> and a <u>mean</u> OR to identify / eliminate / remove anomalies ;	6

Question	Answer	Marks
1(c)	1 axes correctly orientated and labelled ; 2 % / percentage on each axis ; 3 correct line ; 	3
1(d)	<i>two from:</i> 1 muscle strips used are, from a dead animal / <i>in vitro</i> (so response may be different) ; 2 <i>idea that</i> in a living organism muscle contraction is under nervous control ; 3 thickness of the muscle strips used are variable / not testing individual muscle fibres ; 4 <i>idea that:</i> concentration of ATP is not the same as <i>in vivo</i> ;	2
1(e)	<i>two from:</i> 1 <i>idea of:</i> making Ringer's solution(s) with glucose and ATP (and repeating the measurements) ; 2 (then) comparing them with the solutions made with Ringer's solution(s) and ATP ; 3 <i>idea that</i> ATP concentration must be standardised, i.e. the same in both solutions ;	2